

Heads-Up Computing: Opportunities and Challenges of the Next Interaction Paradigm with Wearable Intelligent Assistants

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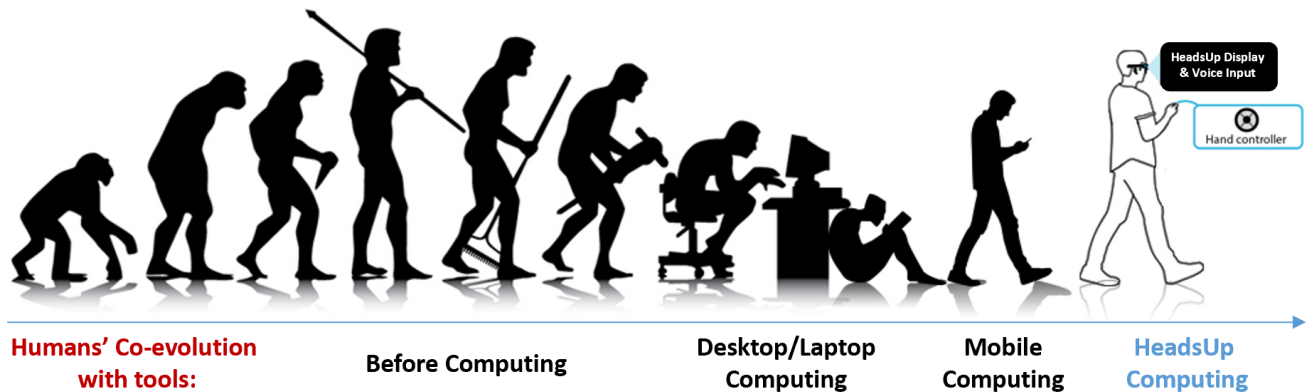


Figure 1: Throughout the extensive history of co-evolution with tools, humans have frequently adjusted their behaviors to accommodate these instruments, as evidenced by the shifts from desktop to laptop to mobile computing. As we chart the course for the next interaction paradigm, how might we draw lessons from past oversights to envision a more harmonious future for human-computer interactions?

Abstract

Modern smartglasses, such as Apple’s Vision Pro and Meta’s Quest series, feature high-end displays and rich multi-modal sensing. They are on the brink of redefining the concept of wearable intelligent assistants. However, despite this unfolding technological revolution, there is much we do not know about how to design for such devices and, ultimately, how they will be used. *Heads-Up Computing* is vision for wearable displays that focuses on how they can enrich human interaction without being obtrusive, on how they can be designed to be both ergonomic and intuitive, and how the interactions that drive them can be aware of, and respond to, a user’s context and resources. Achieving this vision of seamless integration

presents numerous challenges in areas as diverse as providing non-intrusive support, adapting to diverse contexts, enabling expressive input and safeguarding privacy. This workshop aims to convene domain experts to deliberate on these nascent issues and set a collaborative research agenda that prioritizes user-centered design in the ongoing development of Heads-up Computing and seeks to impact the next generation of wearable intelligent assistants.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing; Mixed / augmented reality; Virtual reality.

Keywords

heads-up computing, HMD, AR, MR, VR, Ubicomp, Wearable computing

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1 Introduction

As wearable intelligent devices, such as Apple's Vision Pro [1], Ray-ban Meta Glasses [9], and XReal's Air smartglasses [11], become available to consumers they are inevitably beginning to incorporate them into their daily lives: users have been widely documented (and often derided) wearing them while walking on the street, riding public transit and in a wide range of real world social situations [2]. However, despite the willingness of individuals to experiment with wearing their headsets in diverse contexts, we still know little about designing wearable assistants that can be safely, securely, efficiently and seamlessly integrated into people's daily activities and routines. Indeed, it seems clear that the successful integration of intelligent assistants into everyday activities will, at a minimum, require maintaining a delicate balance between supporting environmental awareness and enabling timely consumption of digital information.

One emerging framework that seeks to address these concerns is "Heads-Up Computing" [12]. At its core, this vision seeks to provide a seamless, timely, and intelligent digital support mechanism, ensuring that technology accentuates human capabilities rather than subduing them. It represents a focused yet flexible interaction paradigm, anchored on three pillars:

- Body-compatible hardware components: devices that conform to and respect human anatomy and ergonomics.
- Multimodal voice, gaze, and gesture interaction: leveraging natural human communication modalities.
- Resource-aware interaction model: recognizing and adapting to the user's current cognitive and physical resource allocation.

Although researchers have started investigating this area [3–8, 10], there is still much to uncover in order to fully understand the big picture. How can wearable intelligent assistants be designed to provide meaningful support without overwhelming or distracting users? How can they adapt to individual contexts and preferences while maintaining privacy and security? These questions highlight the need for a comprehensive exploration of the opportunities and challenges presented by the next interaction paradigm with wearable intelligent assistants.

This workshop aims to bring together experts and stakeholders in the field to collaboratively examine the unique design considerations and implications of wearable intelligent assistants. By analyzing real-world use cases and exploring cutting-edge research, we can gain valuable insights into the potential of these devices to enhance and transform our daily lives. Additionally, we will address the ethical and societal implications surrounding the widespread adoption of wearable intelligent assistants, ensuring that these technologies are developed in a responsible and sustainable manner.

By fostering an inclusive and interdisciplinary dialogue, this workshop seeks to bridge the gap between hardware development and holistic understanding. Through this collective effort, we can shape the future of Heads-Up Computing, not only by addressing the technical challenges but also by considering the human-centered aspects that are critical to the successful integration of wearable intelligent assistants into our lives.

2 Organizers

Shengdong Zhao is a Professor at the City University of Hong Kong. With an extensive background in creating interface tools and applications, Prof. Zhao's research is centered on the development of the next interaction paradigm, referred to as "Heads-up Computing." This endeavor aims to facilitate wearable personal assistants that provide timely, context-sensitive assistance in various environments.

Ian Oakley is a Professor at KAIST with two decades of contributions to wearable technology. His research on smartglasses and unique input methods is integral to *Heads-Up Computing*, and his regular involvement in top HCI conferences illustrates his dedication to wearable interface innovation.

Yun Huang is an Associate Professor at the University of Illinois and co-directs the SALT Lab. Focusing on context in social computing systems, her research is vital for developing context-aware wearable assistants. With a PhD from UC Irvine and a bachelor's from Tsinghua University, Huang's work underpins the human-centric approach in *Heads-Up Computing*.

Haiming Liu is an Associate Professor at the University of Southampton, concentrating on interactive information access, crucial for wearable intelligent assistants. Her expertise in multimodal search interfaces is pivotal for the usability of *Heads-Up Computing* devices. Liu's organizational experience in workshops and her scholarly work in HCI and information retrieval underscore her impact in wearable tech.

Can Liu is an Assistant Professor at the City University of Hong Kong, researching speech-based interfaces and spatial computing for the next wave of wearable assistants. Her work on natural input modalities is enhancing user-friendly *Heads-Up Computing* experiences. Liu's continuous engagement in leading HCI forums marks her as a vital player in the field's progression.

3 Pre workshop plans

The organizers have been meeting semi-regularly in order to prepare the workshop proposal. If the proposal is accepted, we will arrange a formal bi-weekly meeting schedule to discuss issues and define and refine the agenda, activities and outcomes.

To encourage diverse participation, we will adopt a call for submissions with two formats: a detailed position paper or a more concise proposal via our online submission form. Position papers, in line with the ACM Master Article Submission Templates, should be limited to 2 pages, excluding references. These papers can either delve into a case study involving the utilization of wearable intelligent assistants in innovation or research or provide a thoughtful critique of the potential issues of this paradigm. Alternatively, contributors are encouraged to offer insights through our simplified online form. This might particularly appeal to those from interdisciplinary backgrounds or individuals who prefer a more direct method of engagement. With the evolving nature of wearable intelligent assistants, we believe this inclusive approach will facilitate diverse perspectives. We will collect submissions via an online platform such as easychair or HotCRP.

We will advertise our CfP using personal networks, email lists, social media (e.g., X, Facebook) and appropriate aggregator websites (e.g., <http://www.wikicfp.com/>). Our selection criteria for both

articles and text submissions will prioritize both the expertise in wearable intelligent assistants and the relevance of the submission to our workshop's focus. We aim to curate a participant set that reflects a mix of experiences, ensuring a broad spectrum of views and promoting a constructive discussion. In making our selections, we will remain committed to fostering diversity across career stages, racial backgrounds, genders, and regions.

All finalized participants will be integrated into a dedicated digital communication channel (e.g., slack, discord) before the workshop's commencement. This channel will act as an information hub, a space for introductions, a repository for workshop materials, and a collaborative forum for post-event initiatives and discussions.

4 Workshop structure

The one-day **hybrid** workshop is tailored to accommodate up to 20 in-person attendees, with provision for additional online participants. Throughout the day, participants can expect periodic breaks to refresh. Our session leaders, each an expert in their domain, will ensure a deep and engaging discussion. To further enhance the experience, we'll have volunteers or facilitation assistants from our research community on hand, ensuring a harmonious balance between organizers and participants.

4.1 Activity 1: Setting the Stage [30 minutes]

The workshop commences with an elucidation of the "Heads-Up Computing" paradigm. After a concise organizer presentation, we will conduct an ice-breaking activity. Here, participants will be invited to share their thoughts on wearable intelligent assistants.

4.2 Activity 2: Keynote [45 minutes]

We will recruit an invited senior speaker from the research community to offer up their perspective on the workshop themes and issues. This detailed presentation will provide additional perspectives on the core issues the workshop addresses and help set the stage for a lively discussion. Potential speakers we are considering include Mark Billinghurst and Steve Mann.

4.3 15 minutes break

4.4 Activity 3: Participant Research Showcase [45 minutes]

We will provide participants with a platform to elucidate their position statement and perspectives coming into the workshop in a series of succinct 3 minute presentations. The goal will be to present the perspectives attendees bring to the workshop and support fostering a mutual understanding of each attendee's opinions, interests and expertise related to the domain. This activity will help forming discussion groups later in the workshop.

4.5 15 minutes break

4.6 Activity 4: Delineating the Key Themes [30 minutes]

A guided session wherein major themes pertinent to the wearable intelligent assistant domain will be introduced and deliberated upon by the organizers. This serves as a precursor to the more intensive group discussions that follow. The themes will be finalized based on an analysis of the accepted workshop contributions (i.e., they

will reflect the content submitted by attendees). The themes may include some or all of the following:

- **Interfaces and Interactions** : As smart glasses usher us into a new age, they bring forth the question of designing interactions that are intuitive, seamless, and socially acceptable. How can we meld technology with human instincts?
- **Mobility/Multitasking** : The mobility that smart glasses bring is undeniable. The design nuances of catering to a user on-the-move—be it walking, driving, or merely existing in public spaces—deserves detailed discussion.
- **Ergonomics and Comfort** : Functionality does not necessarily warrant comfort. Balancing capability with user comfort will be a pivotal area of exploration.
- **Inclusive and trustworthy Information Access** : Information empowers people's lives. With a constant influx of information, users stand at the risk of being overwhelmed. This theme will dissect the impact of information accessibility and how to manage and interact with information without jeopardizing safety.
- **Privacy and Ethics** : In an age where user data is holds high value to various holdholders, wearable technologies walk a fine line between being informative and invasive. The ethical implications of data collection, storage, and usage will be a prime area of focus.
- **Abuse and Addiction** : Every technological marvel comes with its own set of pitfalls. The potential misuse, both by vendors and individuals, will be scrutinized. Delving into these dark patterns will help us forecast and possibly prevent misuse.

4.7 Lunch Break

We will arrange lunch at an external restaurant with all workshop participants. If the workshop is accepted, we will attempt to acquire sponsors to support this meal cost (e.g., from a relevant industry partner).

4.8 Activity 5: Group Deliberations round 1 [60 minutes]

Participants will fragment into smaller, focused breakout groups. Initial group membership will be informed by the perspectives shared in the participant research showcase. It is anticipated that each group will gravitate towards one or more of the topics introduced as key themes, engaging in profound discussions about the underlying challenges, opportunities, and implications of each. Each group will be moderated by either a workshop organiser or a nominated senior member. The role of this individual will be to keep the discussion on track. In addition, in the closing 15 minutes of this session, the moderator's role will be to informally document key outcomes and ideas from the discussion (e.g., by creating an affinity diagram of post-it notes or a hand-drawn poster).

4.9 15 minutes break

4.10 Activity 6: Group Deliberations round 2 [60 minutes]

Groups will be merged into larger units, most likely by merging pairs of groups. Each new group will briefly introduce the outcomes of their small group discussions then move on to a wider debate

about how the different topics they have engaged with relate to or impact one another. The point of this activity is to bring groups out of any silos they may have fallen into and engage them with broader perspectives.

4.11 15 minutes break

4.12 Activity 8: Reconvene and Wrap-up [45 minutes]

The final set of groups will surface their discussions, outlining:

- Elucidation of their group vision for wearable intelligent assistants.
- Innovative solutions and suggestions.
- Foreseen challenges and countervailing strategies.
- Impacts and implications of their deliberations.

Inter-group discussions will then ensue, allowing for development of richer and more diversified viewpoints.

4.13 Activity 9: Way Forward [30 minutes]

In the concluding act, the discourse will pivot to the overarching themes of Heads-Up Computing. Questions around its viability, emergent challenges, and the road-map to innovation will steer this discussion. The collective insights will subsequently frame the post-workshop initiatives and solidify the research trajectory for this nascent interaction paradigm.

5 Asynchronous Engagement

Recognizing the diverse needs of our attendees and the unpredictability of technical or accessibility challenges, we've developed strategies for asynchronous engagement. All workshop materials, including session recordings, transcripts, slides, and supplemental resources, will be uploaded to a dedicated section on our workshop website. Furthermore, to ensure a collaborative and interactive experience even post-session, these materials will also be made accessible via shared Google Drive, enabling participants to comment, annotate, and discuss content at their convenience. To foster ongoing conversation and knowledge exchange across different time zones, we will also establish dedicated communication channels on platforms like slack or discord. This approach not only allows for discussions to evolve asynchronously but also ensures that every participant, regardless of their geographical location or access constraints, remains integrated into the workshop's community and discourse. All organizers have committed to supporting and monitoring these channels to address queries, feedback, or further discussions, ensuring a holistic and inclusive workshop experience for all.

6 Post-Workshop Commitment

We pledge to keep the momentum going post-workshop. Our endeavors will include:

- **Webinars and Virtual Meet-Ups:** Regular virtual events will ensure the community stays connected, discussing the latest advancements and challenges.
- **Resource Portal:** A dedicated website will host resources, research papers, discussions, and forum threads, ensuring continuous learning and collaboration.

- **Collaborative Projects:** Encouraging workshop participants to undertake collaborative projects will be a primary focus, ensuring the ideas discussed don't just remain on paper.
- **Future Events:** We will explore options for future workshops at CHI or related venues (e.g., UbiComp). We will also explore the possibility for longer workshop events, such as by organizing a Dagstuhl seminar.

7 Plans to Publish Workshop Proceedings

Documenting our progress is essential. Every insight, discovery, and hypothesis in the field of wearable intelligent assistants is a foundation for what comes next. To keep these contributions within reach, we will compile a detailed proceedings document. This will include all accepted position papers, which will be available on the ArXiv platform for broad accessibility.

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